

MR SAFETY CONSIDERATIONS

Below is a list of MR considerations that the screening form does not explicitly describe and participants often forget to report.

PARTICIPANT CHARACTERISTICS

HEIGHT AND WEIGHT

The scanner bed has a weight limit of about 450lbs, but a large participant will have difficulty fitting comfortably into the machine long before that limit is reached. Your scan team will get precise height and weight measurements the day of the scan for registration, but lab personnel should get at least an approximation of the person's size beforehand.

CCN does not stipulate a specific height/weight limit because the way individual bodies distribute weight is a significant factor in determining their suitability. Generally, however, be cautious with people over 6 feet and 200+ lbs. Reach out to ccnsupport@g.ucla.edu if you would like to bring in a potential participant and try out the fit in the mock scanner. For reference, the diameter of the scanner bore is 22in, the width of the cushioned table area is 19in, and the length of the cushioned area is 75in (including 32ch head coil).

METALWORKING

If someone does welding or other metalwork habitually, even if they wear eye protection and have no history of accidents, they cannot be scanned. Under clinical radiology protocols, these people would be sent for x-ray orbitals to verify no metal in the eyes. CCN has no capability to order these, so we cannot take the risk. The concern is that microscopic metallic particles can build up in the eyes and/or on the skin over time, causing burns if exposed to the field for extended durations.

The same applies to old eye injuries involving metal--unless the participant can show evidence of MR safety in the form of ocular x-rays, they cannot be scanned. Recent documented clinical 3T head MRIs can be taken into consideration, but do not guarantee clearance.

COMMON METAL & MEDICAL PROCEDURES

DENTAL WORK

Always ask about all dental work. It is very easy for people to forget about common fixtures such as fillings, braces, and retainers. Most dental implants are safe, but devices like wires and palate expanders are exceptions. Since they need to be flexible, they are often made of ferrous materials. Even if you are certain the participant's dental work does not pose a safety risk, extensive dental work can lead to imaging artifacts. There is no specific guidance other than that more dental work entails higher risk of artifact.

IUD

Implanted contraceptives are very common. Frequently-seen brands include Mirena, Kyleena, Paragard (IUDs) and Nexplanon (arm implant). These devices are often cleared, but still need to be assessed on an individual basis. Different variants from the same brand may have different conditionality and CCN needs to check these specifications every time, especially if your participant has other implants as well.

RECENT SURGERIES

Compile information about surgical procedures performed within the past ~12 months. A recent surgery is not necessarily a contraindication, especially if no implants were involved, but details are required to get a full picture of the situation. Some procedures come with post-surgical recommendations because materials need to be removed after a certain amount of time has passed. There may also be concerns related to the (sub-)acute recovery period, such as side effects of medication, discomfort lying still for a long time, and potentially some amount of implant migration as it gets settled into the tissue. The typical recommendation is to wait at least 6 weeks post-surgery to allow for proper wound healing.

PROCEDURES AND ITEMS THAT “SOUND SAFE”

"Any metallic or foreign body" is listed on the screening form, but people will often overlook items that apply to them if they've gotten used to it or assume it doesn't count as metal. Make sure you follow up on injuries, procedures, and foreign objects, even if they appear on first pass to involve no metal. For example, a ceramic shard embedded in the skin from a recent injury may not be ferrous itself, but could be painted with a metallic glaze and therefore susceptible to heating up.

ACCESSORIES

WIGS, WEAVES, HAIRPIECES, CLIPS, AND EXTENSIONS

Almost all of these will cause significant artifact and, in some circumstances, heat up. Participants must remove all hair fastenings or they can't be scanned. For the sake of time and simplicity, you may want to ask all participants to refrain from wearing anything in their hair and applying any specialty products on the day of their appointment.

Screen carefully prior to allowing the participant into the control room. Have them run their hands through their hair in front of your scan team. Some hair accessories are not visually obvious and can be missed by the wand if the user doesn't push it close enough to the head. These items include bobby pins, clips, and extensions.

IRREMOVABLE JEWELRY

The form mentions jewelry, but make sure you are not only checking for rings and piercings. Permanent jewelry like "forever bracelets" that can't be removed have become very popular. These items don't absolutely preclude someone from participation, but you will definitely need information about the material, size, and location ahead of time.

NAIL ART

Nail art can be very ornate and involve special materials that aren't immediately obvious. Magnetic nail polish (sometimes called a "cat eye" design) has become popular for its color-manipulation properties. It is prohibited from the scanning environment because it may cause burns to the nail beds. CCN stocks nail polish remover in the control room.

More involved nail decor (sequins, foil, glitter, gems, etc.) may also pose a heating risk. Moreover, flakes and other small particles could get into the MR system and cause artifacts, resulting in extensive CCN downtime. Advise your participants of this risk well ahead of time so they can plan accordingly. Participants are typically willing to wipe off polish on short notice, but unwilling or unable to remove elaborate nail art that was done at a professional salon.

COSMETICS

TATTOOS

Tattoos are extremely common and present different levels of risk depending on a number of factors:

- **Location:** Tattoos of any kind on the face or head are strictly prohibited. Neck tattoos can be considered based on other factors. The further from the head, the lower the risk.
- **Professionalism:** Tattoos that were not done by a professional in a salon are prohibited. Tattoos done outside of this setting--stick-and-poke jobs at home, in prison, etc.--are not permitted because there is no way of knowing the composition of the ink that was used.
- **Origin:** "Professionally done in the US" gives CCN the best idea of the standards and materials involved. Tattoos done outside the US are not necessarily exclusionary, but include this information when requesting clearance.
- **Size:** The larger the inked area, the greater the risk.
- **Age:** Ink is more concentrated in fresh tattoos and diffused in older ones.
- **Color:** Iron oxide is particularly common in pigments for red and black ink.

Request CCN approval for all but the clearest of safe cases (i.e., a tiny professional ankle tattoo). Radiological planning tattoo markers (used in radiation therapies for cancers) may be scanned.

PERMANENT MAKEUP

The screening form lists permanent makeup, but participants often forget about cosmetics they have been living with for years. As with tattoos, permanent makeup on the face or head immediately excludes your participant from scanning. Anything near the eyes in particular (e.g., permanent eyeliner) is highly risky and will not be considered.

The only exception is microbladed eyebrows, on the condition that at least 6 weeks have passed since the procedure. Eyebrow microblading has been extensively documented and shown to be safe (with monitoring). You may find it useful to explicitly cite microblading and permanent eyeliner as examples to help trigger your participant's memory during initial screening.

REMOVABLE MAKEUP & COSMETICS

It is recommended that you ask the participant to come in for their scan wearing no makeup at all. Even products like specialty mascara may contain tiny metallic fragments or pigments that cause reactions around the eyes and/or artifact on the scan.

Eyelash extensions are very common:

Glued: Must be removed and glue residue washed off.

Magnetic: Magnetic extensions and all related products must be completely removed PRIOR to arriving at the imaging center. These extensions are attached to the eyelid using tiny magnets and/or a magnetic eyeliner that is applied before the lashes clip on. Removing the lashes after arrival at CCN is not sufficient. Everything must be removed and washed off before the participant comes in. Magnetic nanoparticles from these products is not only hazardous to the participant, but could also potentially contaminate the scanner environment.

As beauty products continue to develop and fall in and out of fashion, the composition of any given cream, powder, gloss, or other substance becomes more unpredictable. It is safest and simplest to request no makeup at all, including cosmetics that are glued or clipped on. CCN stocks makeup remover wipes in the control room.

DEODORANT & PERSONAL CARE PRODUCTS

Antiperspirants often contain trace amounts of aluminum. Similar to the guidance regarding hair and makeup, it is safest to ask your participant to refrain from applying deodorant, lotion, perfume, or hair products prior to their appointment. These items may contribute to skin warming and/or image distortion depending on their ingredients.

COLORLED CONTACT LENSES

Cosmetic contact lenses of any kind--such as colored, circle, ring or big eye--are prohibited. These lenses may contain iron oxide and/or other materials to create the colored effect. Participants may only wear normal, uncolored, prescription contact lenses while scanning. If you are unsure about the type of contacts your participant is wearing, they should be removed.

CLOTHING

ATHLETIC WEAR

Athletic clothing is sometimes made with metallic fibers that help imbue the material with antimicrobial or odor protection properties. X-static nylon is one such component. This is silver bonded to nylon that is prevalent in (but not limited to) Lululemon products and yoga pants. Be aware that not all clothing manufacturers know the source of their materials for athletic wear.

Tight clothing is less preferred for MRI in general for temperature regulation. Advise your participant to wear loose-fitting, plain cotton clothes for their scan. CCN stocks scrubs in children's (M) and adult's (M, XL, 3XL) sizes.

MENSTRUAL UNDERWEAR

Similar to the concern with athletic wear, menstrual underwear (e.g., Thinx) sometimes contains metallic microfibers as antimicrobial/anti-odor agents. Consider advising your participant to use disposable products with plain cotton undergarments instead.

POCKETS

Advise your participant that pockets with metal buttons are not permitted and pockets in general should be minimized (ideally absent). Every pocket is a potential vector of projectiles being carried into the magnetic field--this is true for staff as well as participants.

If the participant arrives wearing an outfit with pockets, have them change into scrubs, or at the very least, physically turn out every pocket in front of lab personnel. Reaching into pockets and/or verbally attesting that they are empty is not sufficient.